

COLLEGE OF DEFENCE MANAGEMENT · DATATHON - 2026

Theme: Global Conflicts – Impact on Supply Chains

PROJECT DARKWATER

Global Conflicts and the Blinding of Supply Chains: Maritime Picture Assurance at Contested Chokepoints, with Implications for India and the Indian Armed Forces

Commander's Edition

What the 2026 war did to the ships that carry India's oil and trade, what it cost, how long such crises last, and what India and its Armed Forces should do. Told as one story, in plain words.

Submitted by

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[Service No] · [Unit / Formation]

September 2026

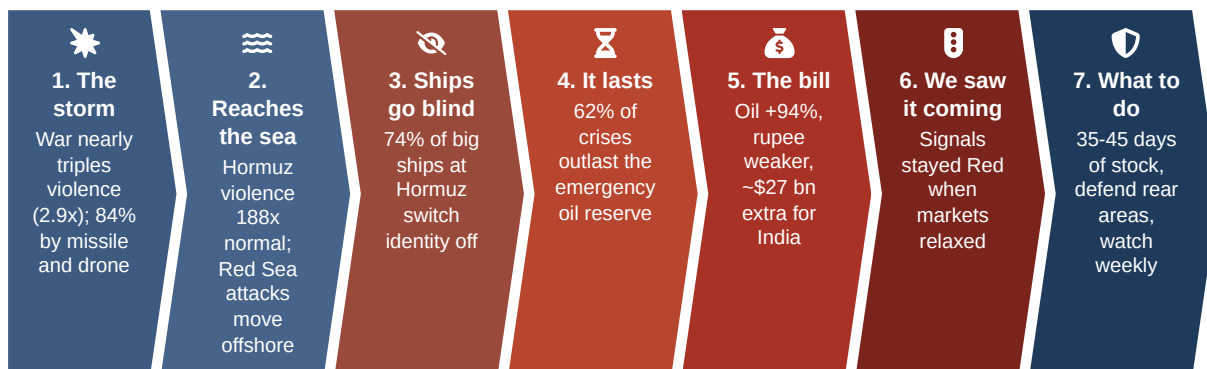
Companion documents: full technical report, Power BI dashboard (.pbix), analysis code and data

THE STUDY AT A GLANCE

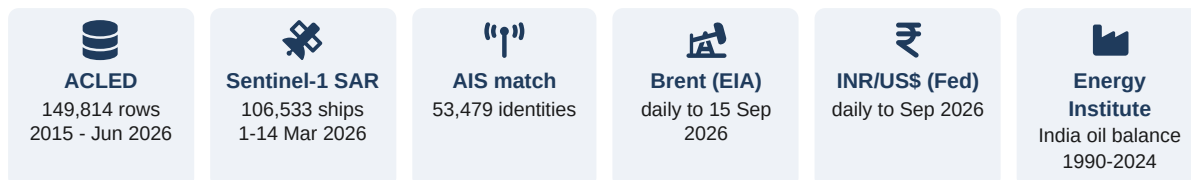
PROJECT DARKWATER · Global Conflicts and the Blinding of Supply Chains

CDM Datathon-2026 · Theme: Global Conflicts - Impact on Supply Chains · the study on one page

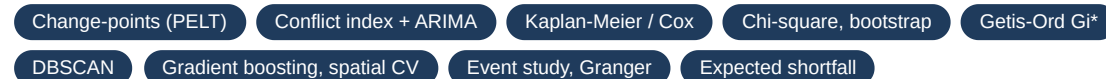
THE STORY IN SEVEN STEPS



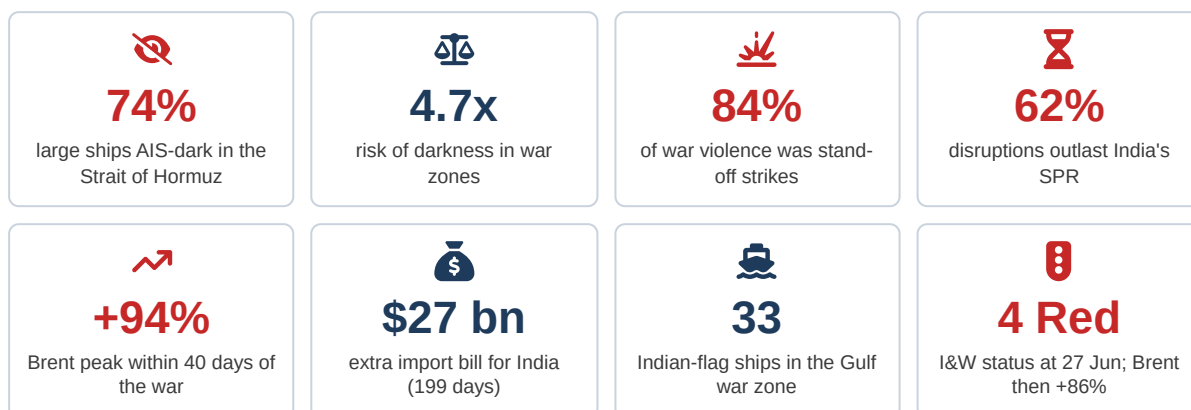
DATA FUSED



METHODS



WHAT THE DATA SAYS



Bottom line: war at a chokepoint first **blinds** the supply chain, then **outlasts** the buffers. Hold 35-45 days of cover, harden rear areas against stand-off strikes, and watch the chokepoints weekly with data.

EXECUTIVE SUMMARY

BOTTOM LINE UP FRONT. When war reaches a sea chokepoint, the first thing to fail is the **maritime picture**: big ships stop identifying themselves, so AIS-based counts overstate how much shipping has gone and hide what is still there. The disruption then **outlasts our reserves** and cost India about **US\$27 billion** in six months. Our warning signals saw it coming when the market did not.

Situation

War reached the Strait of Hormuz on 28 February 2026. This study joins 11 years of conflict data, satellite radar of 106,533 ships in the first fortnight, and open oil, rupee and energy data to September 2026.

Assessment

1. **The storm.** Violence nearly tripled (2.9x); 84% by missiles and drones; the Gulf states were hit 124 times more than before.
2. **The picture failed.** 74% of big ships at Hormuz were dark against 10% in peaceful seas. AIS counts fell 83%, but radar saw most hulls still there, many of them held at anchor. The Gulf darkened 5.7 points a day faster than 11 control seas.
3. **Four behaviours, not one.** Concealment in the Gulf, evacuation from the Red Sea, a frozen Black Sea, and compliance in the East Med (8% dark beside heavy fighting).
4. **It lasts.** 62% of sea-lane crises outlast India's ~10-day emergency oil reserve; about 1 in 10 outlasts total national cover.
5. **The bill.** Brent \$71 to \$138; in March the extra oil bill equalled 59-70% of the monthly defence budget.
6. **We saw it coming.** On 26 June the market relaxed (\$70); our DARKWATCH signals stayed Red. Oil then rose 86%.

Deduction

The decision variable is the **state of the corridor**, not AIS silence alone. Classify first (concealment, evacuation, frozen, compliance), then act. Each needs a different response: escort and surveillance, rerouting, provenance checks, or sustained presence.

Recommendation

- Make **Maritime Picture Assurance** a named joint warning function under HQ IDS, run through IFC-IOR, with the SDR and DARKWATCH as standing products.
- Size fuel, aviation-fuel and critical-spares reserves for all three Services at **35-45 days**; harden rear areas against drones and missiles; GPS-resilient navigation.
- Standing rule: **no AIS-derived traffic count** for a contested corridor without a radar cross-check.

Decision sought

Approve a **four-month Phase 1 proof of concept** using existing establishment, free Copernicus Sentinel-1 imagery and the analysis pipeline already built. Continue only if a principal acts on a DARKWATCH warning in that period.

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Each chapter opens with **The story so far** and closes with **So what**; every chart carries **What this shows**. A commander can read the Executive Summary and the coloured boxes alone in about ten minutes.

1. THE QUESTION AND HOW WE ANSWERED IT

PROLOGUE - THE QUESTION

THE STORY SO FAR

On 28 February 2026 war reached the shores of the Strait of Hormuz, the passage for about a fifth of the world's oil and a large part of India's. This edition answers five questions a commander would ask: what happened, what did it do to the ships, what did it cost, how sure are we, and what should we do?

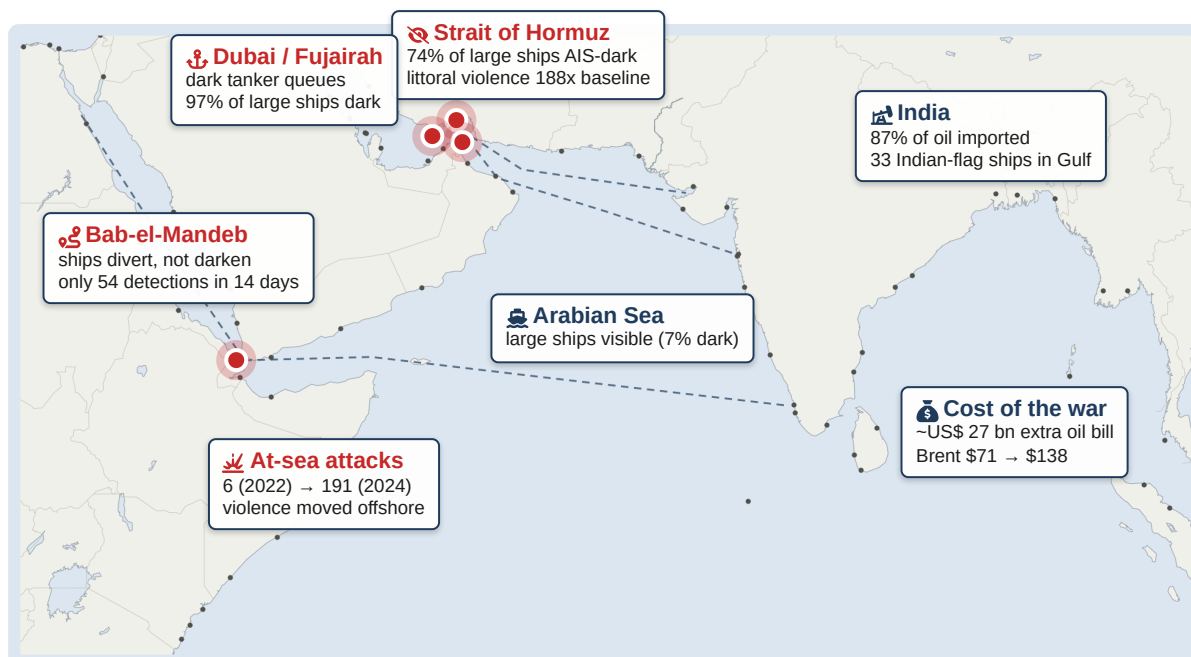


Figure 1.1 : The theatre: India's western sea lanes, the chokepoints and the headline findings



Figure 1.2 : How the question was answered

The two CDM datasets are the **ACLED** conflict record (every reported attack, battle and protest in the Middle East, week by week, 2015 to June 2026) and **Sentinel-1 satellite radar** detections of 106,533 ships in the first fortnight of the war, each checked against the ship's identity beacon (AIS). Open-source oil-price, exchange-rate and energy data (to September 2026) measure the cost. The glossary at the end explains every technique in plain words.

SO WHAT

Two datasets from CDM (11 years of conflict events; 106,533 ships seen by satellite radar) plus four open-source datasets (oil price, rupee, India's oil balance, world ports), tested with standard analytics and intelligence tradecraft.

Next: How did the conflict itself change?

2. THE STORM GATHERS

PART I - THE STORM GATHERS

THE STORY SO FAR

The story begins on land. Before the war reached the sea, it changed in size, in weapons and in geography.

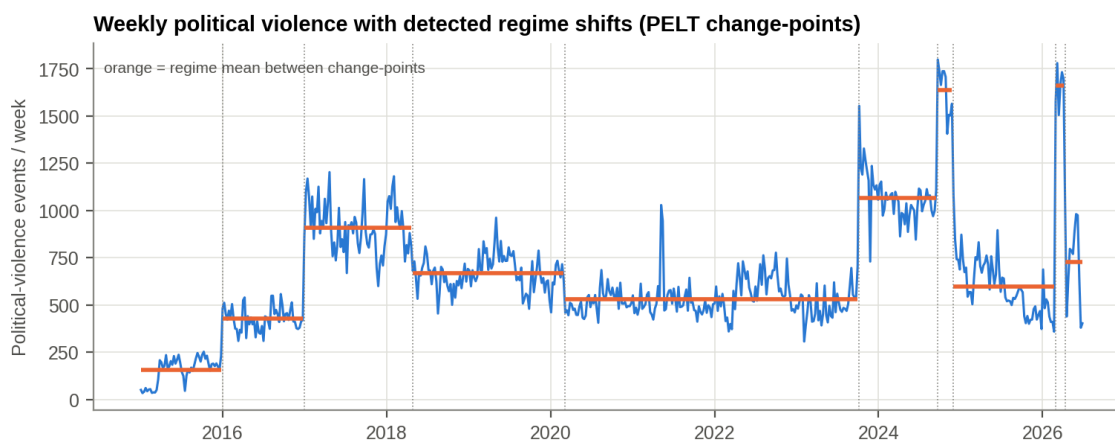


Figure 2.1 : Weekly political violence, 2015-2026, with the dates the 'battle rhythm' changed

What this shows: The orange steps are the 'battle rhythm'. The computer found, without being told, the exact weeks it changed: 7 Oct 2023 and 28 Feb 2026.

Without being told any dates, the computer found the weeks when the conflict changed: **7 October 2023** (the Gaza war) and **28 February 2026** (the regional war). In the war weeks violence ran at **2.9 times** the previous year's level. **84%** of it was delivered from a distance, by missiles, drones and artillery.

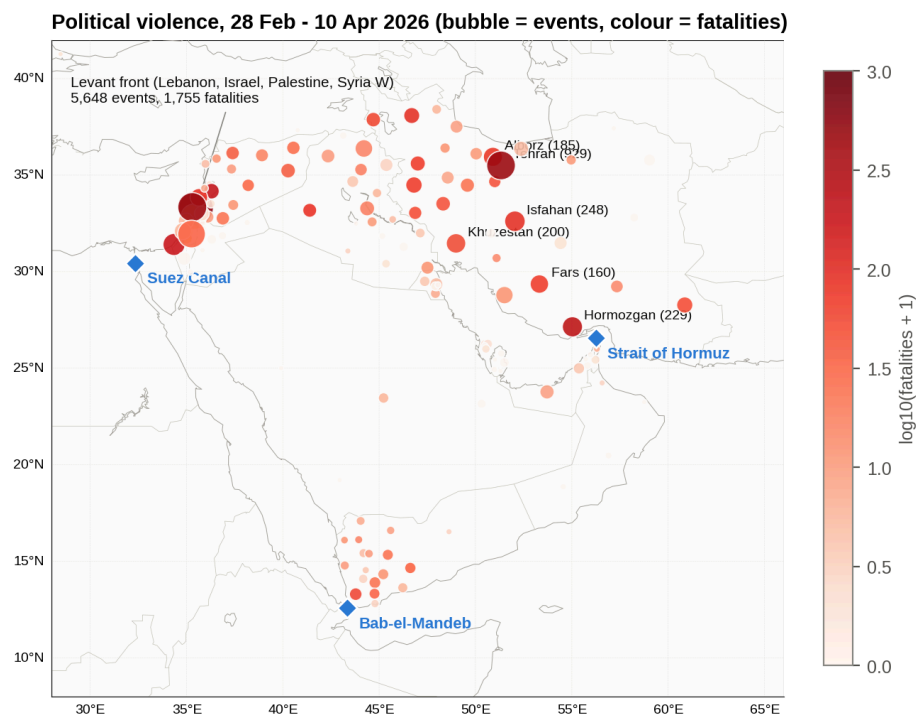


Figure 2.2 : Where the violence fell, 28 February - 10 April 2026

What this shows: Each bubble is a province: bigger means more attacks, redder means more deaths. The fighting reached the shore of the Strait of Hormuz.

The fighting reached the **shore of the Strait of Hormuz** (Hormozgan province: 229 attacks, 215 deaths in six weeks). The Gulf states, which had seen almost no violence, were struck every week: **124 times** their pre-war rate.

SO WHAT

The war nearly tripled violence (2.9x). 84% of it was delivered by missiles and drones, and it hit the Gulf states that export the world's energy (124x).

Next: Did it reach the sea lanes?

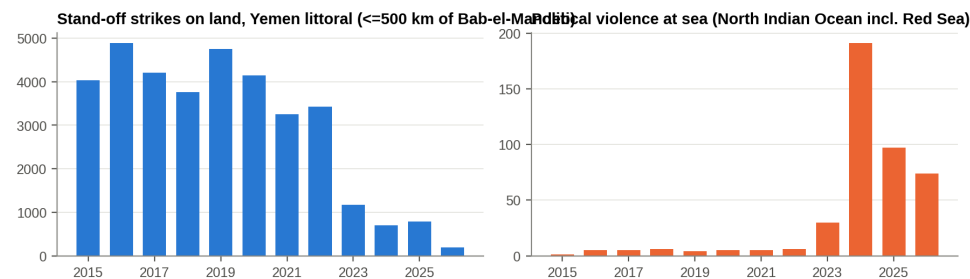
3. THE STORM REACHES THE SEA LANES

PART II - THE STORM REACHES THE SEA LANES

THE STORY SO FAR

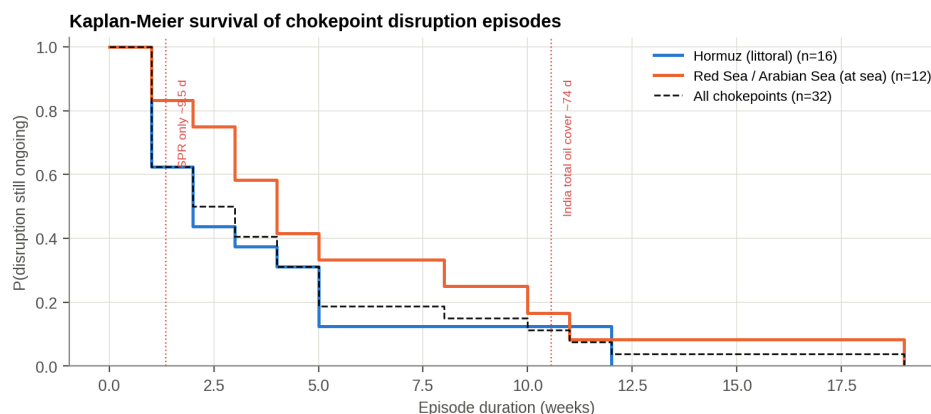
The war had moved to the Gulf's energy coast and was being fought with missiles and drones. The next question: did it reach the sea lanes, and how long do such crises last?

Red Sea theatre - the violence moved offshore after 2023

**Figure 3.1** : Red Sea: attacks on land fell while attacks at sea rose

What this shows: In Yemen, attacks on land fell (left) while attacks at sea rose (right). The threat moved from the shore to the shipping lanes.

A 'thermometer' of violence was built for each sea route. Around Hormuz it reached **188 times** its normal level. In the Red Sea the threat **moved offshore**: attacks at sea rose from 6 (2022) to 191 (2024) while the land war faded. A land-only measure would have shown the Red Sea getting safer just as it became the most dangerous waterway for merchant ships.

**Figure 3.2** : How long do sea-lane crises last? Share still going after each week (red lines = India's oil reserves)

What this shows: Read it like a survival curve: the line shows the share of crises still going after a given number of weeks. The red lines are India's oil reserves.

Most crises end within weeks (median 2.5 weeks), but some run for months (longest 19 weeks). **62% outlast India's strategic petroleum reserve** (about 9.5 days) and about 11% outlast total national cover (about 74 days). The danger is not the spike but its duration.

SO WHAT

Yes. Violence around Hormuz reached 188x its normal level, and in the Red Sea it moved from shore to sea. Most crises end in weeks, but 62% outlast India's emergency oil reserve.

Next: What did the ships do?

4. THE SHIPS GO BLIND

PART III - THE SHIPS GO BLIND

THE STORY SO FAR

Violence reached the waters around Hormuz and the Red Sea, and such crises can outlast our reserves. What did the ships do? Satellite radar sees every ship, whether or not it wants to be seen.

AIS-dark share by sea region - orange: war zones, violet: India's seas

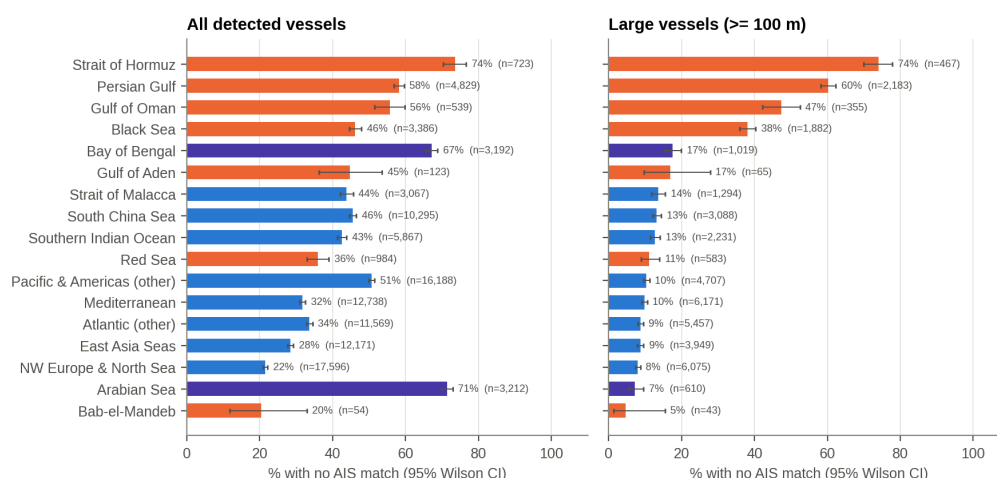


Figure 4.1 : Share of ships with their identity beacon off, by sea (right panel: big ships only)

What this shows: Look at the right-hand panel (big ships only). War zones (orange) are four to seven times darker than peaceful seas (blue).

Small boats run without beacons everywhere, so the test that matters is **big ships (100 m and over)**, which must carry one by law. In peaceful seas about 1 in 10 is dark. At **Hormuz it was 74%**, in the Persian Gulf 60%, and in the Black Sea, the other war zone, 38%. Big ships in war zones were **4.7 times** as likely to be dark.

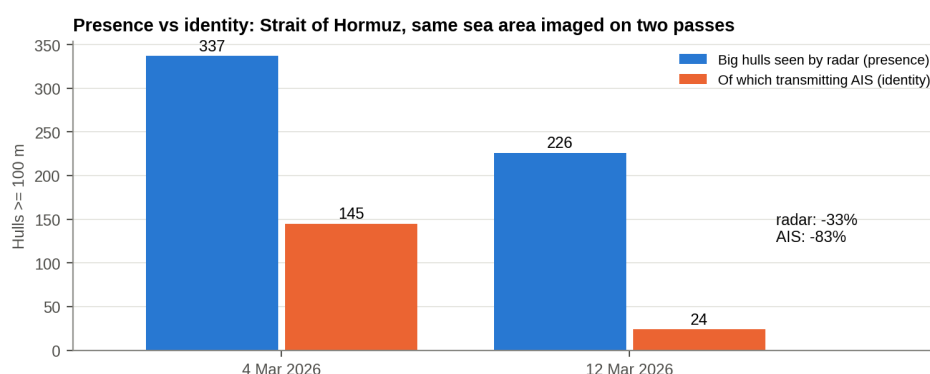


Figure 4.2 : Did the ships leave or go silent? Same sea area at Hormuz, imaged on 4 and 12 March

The ships did not simply leave. On the same patch of sea, AIS-transmitting big hulls fell 83% while radar saw the physical fleet fall only 33%, and the dark fleet did not shrink at all. Many dark hulls were **held at anchor**: seen in the same 100 m square on different days far more than chance allows ($z = 5.3$). An AIS-based traffic count therefore overstated the collapse of shipping and hid the collapse of the maritime picture.

A master switches off AIS when being identified is dangerous **and** not sailing is not an option. Read the wrong signature and you get the wrong response.

🚫 CONCEALMENT

Identity dangerous · route cannot be given up

Measured: Hormuz SDR 74%, Persian Gulf 60%, rising ~6 pts/day

Supply chain: Price and insurance shock with cargo still moving; picture collapses

Staff response: Non-AIS surveillance, escort, war-risk cover; do NOT assume cargo stopped

🚢 EVACUATION

Identity dangerous · route can be given up

Measured: Red Sea SDR 11%, Bab-el-Mandeb/Aden 12%; traffic thins

Supply chain: Volume, transit-time and freight shock system-wide

Staff response: Re-plan lead-times via the Cape; inventory and port buffers

⌚ ATTRITION / FROZEN

Long war · two fleets settle in

Measured: Black Sea SDR 38%, flat trend

Supply chain: Persistent premium; sanctions and provenance risk

Staff response: Pattern-of-life baselining; provenance checks

🛡️ DETERRENCE / COMPLIANCE

Identity protective · authority enforces reporting

Measured: East Med / Suez SDR 8% with 1,497 violent events within 300 km

Supply chain: Largely unaffected: what a protected sea line looks like

Staff response: The benchmark Indian presence should aim to produce

Figure 4.3 : Four behaviours at sea: the same war produced concealment, evacuation, a frozen sea and compliance

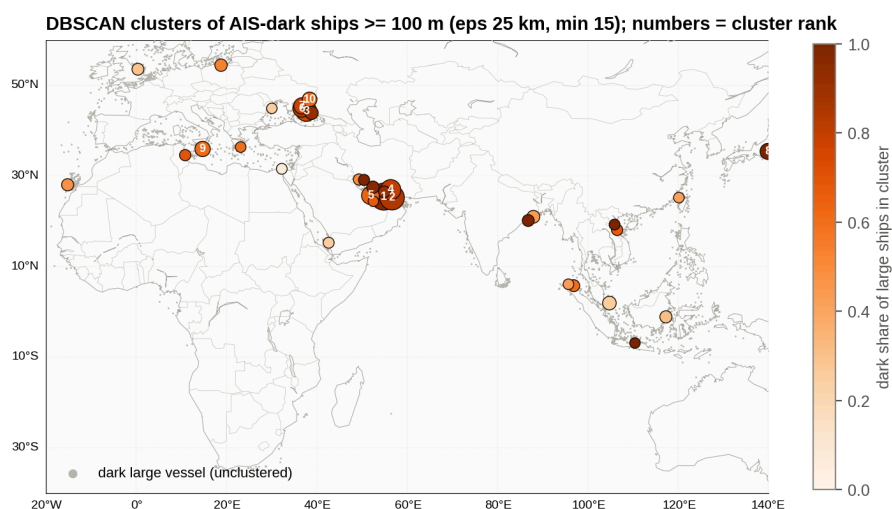


Figure 4.4 : Where dark ships gather: tanker queues off Dubai and Fujairah, and in the Black Sea

What this shows: Each circle is a group of dark ships found automatically. The biggest are tanker queues off Dubai and Fujairah, waiting with identities off.

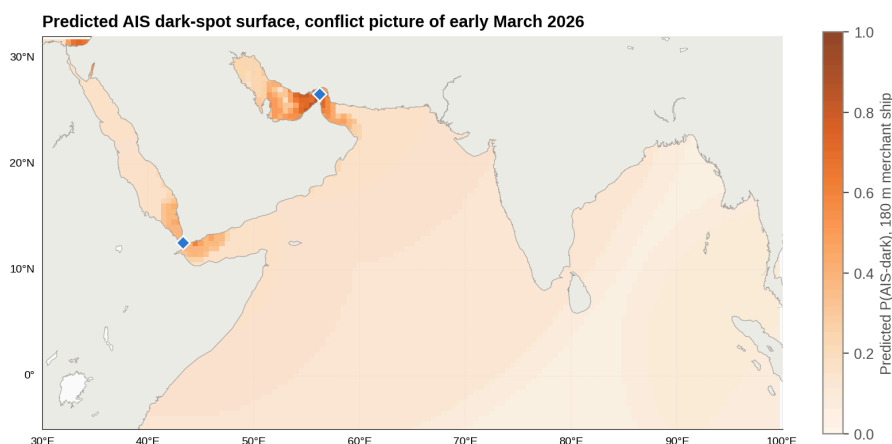


Figure 4.5 : Forecast: where a 180 m merchant ship is likely to go dark

What this shows: A forecast map of where a merchant ship is likely to go dark. Darkest near Hormuz and the UAE coast, lighter near India.

A machine-learning model, trained on 62,000 ships and tested only on seas it had never seen, forecasts where ships will go dark (score 0.69, where 0.5 is a coin toss). After a ship's size, **distance from the fighting** is the strongest clue. It is good enough to point satellites at the right waters.

SO WHAT

Big ships went dark: 74% at Hormuz against about 10% in peaceful seas. They did not simply leave; many were held at anchor with identities off. The same war produced four behaviours at sea, and a model can forecast where darkness will appear.

Next: What did it cost?

5. THE BILL ARRIVES

PART IV - THE BILL ARRIVES

THE STORY SO FAR

Ships near the fighting switched off their identities or stayed away. A supply chain is ultimately about goods and money, so we follow the shock to the oil price, the rupee and India's import bill.

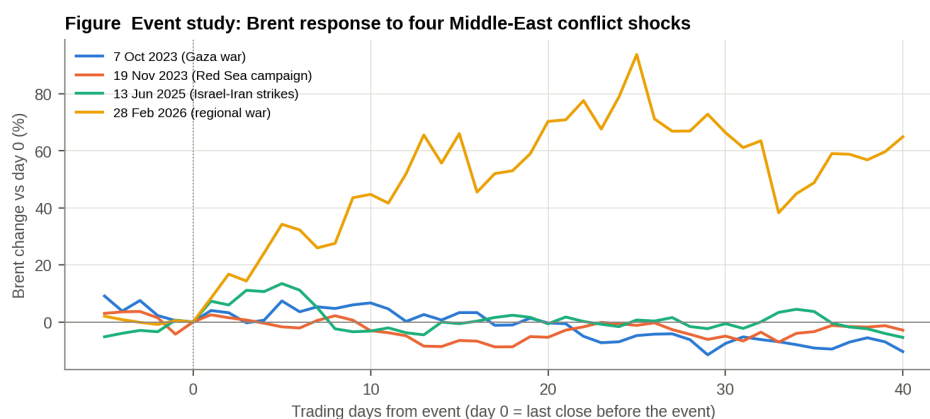


Figure 5.1 : Oil price after four Middle-East shocks (day 0 = the day before each began)

What this shows: Each line is one crisis, starting at zero on the day it began. Only the 2026 war (red) produced a lasting jump.

Only the war that reached Hormuz produced a lasting shock: **Brent +94%** within 40 days, from \$71 to a peak of \$138. The Red Sea campaign, where ships could detour, left oil *lower* a month later. The market confirms the blinding-versus-diversion finding.

Table 5.1 : What the war cost India (open-source data)

Quantity	Value
India net oil import volume (2024, OWID-derived)	4.26 million bbl/day
India oil import dependence (2024)	87.4%
Brent, pre-war baseline (1 Dec 2025 - 27 Feb 2026)	US\$ 66.6/bbl
Brent, war period (1 Mar - 15 Sep 2026)	US\$ 98.4/bbl (+48%)
INR/US\$, pre-war vs war period	90.54 -> 94.72 (+4.6%)
Extra import bill, war period (199 days), US\$	~US\$ 27 billion
Extra import bill incl. rupee effect, INR	~Rs 2.8 lakh crore
Sensitivity: every US\$10/bbl sustained for a year	~US\$ 15.6 billion

India imports **87%** of its oil, and the share is rising. The war added about **US\$27 billion (Rs 2.8 lakh crore)** to the oil bill in 199 days. Every US\$10 a barrel for a year costs about US\$15.6 billion.

In defence terms: in March 2026 the extra oil bill equalled **59-70% of the whole monthly defence budget**; in April it reached 100%.

Could we have seen it coming? On 26 June 2026, when the conflict data end, the oil market looked calm: Brent was back at \$70. The six warning signals built in this study all read Red or Amber and said *hold the buffers*. Oil then rose **86%** to \$131 by 15 Sep 2026. The data saw what the market missed.

SO WHAT

Oil went from \$71 to \$138, the rupee weakened, and India's oil bill rose by about US\$27 billion. Our warning signals stayed Red when the market relaxed, and oil then rose 86%.

Next: How sure are we?

6. INFERENCE: WHAT WE KNOW AND HOW SURE WE ARE

PART V - WHAT WE KNOW, AND HOW SURE WE ARE

THE STORY SO FAR

The evidence is complete. Before acting on it, a good staff officer asks: are we sure, and could something else explain it?

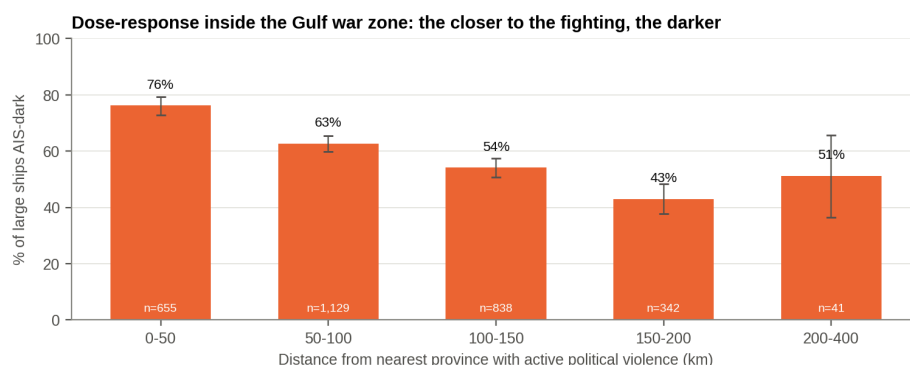


Figure 6.1 : The closer to the fighting, the more big ships went dark (Gulf war zone)

What this shows: The closer ships were to the fighting, the more of them went dark: a clear sign that the war caused it.

Four tests make the main finding solid. (1) It holds under **eight different ways** of measuring it (war zones 3.9 to 6.4 times darker). (2) It gets **stronger nearer the fighting**, from 76% within 50 km to 43% at 150-200 km. (3) It **repeats in a separate war** (Black Sea). (4) Of four possible explanations (deliberate switch-off, GPS jamming, radio reception gaps, computer error), only **deliberate switch-off** fits all the evidence. (5) It survives removing the 67 passes that were entirely dark (a possible feed outage) and holds by day and by night. (6) The Gulf darkened 5.7 points a day faster than 11 control seas ($p = 0.008$). A failed identity match proves the picture failed; it cannot prove intent, so that judgement is held at 'likely'.

Table 6.1 : Key judgements (intelligence estimative language)

How likely	Judgement	Confidence
Almost certain	The war caused big ships to go dark.	High
Likely	Most of the darkness is deliberate switch-off; GPS jamming probably adds to it.	Moderate
Highly likely	The Hormuz coast stays in crisis, at least on and off, through September 2026.	Moderate
Likely	A sea-lane crisis outlasts India's emergency oil reserve (62% of cases).	Moderate
Almost certain	Drones and missiles will remain the main threat to rear-area logistics and energy sites.	High
Highly likely	Conflict and satellite signals warn earlier than market prices.	Moderate

SO WHAT

Very sure of the main finding. It holds however it is measured, grows stronger closer to the fighting, repeats in another war, and no other explanation fits.

Next: What does it mean for the world and for India?

7. WHAT IT MEANS FOR THE WORLD AND FOR INDIA

PART VI - WHAT IT MEANS FOR THE WORLD AND FOR INDIA

THE STORY SO FAR

Having established what we know and how sure we are, we turn to what it means: first for the world, then for India.

For the world

- **Energy:** about a fifth of the world's oil passes Hormuz. Chokepoint risk is now a recurring cost, not a rare event.
- **Trade:** where a detour exists, conflict adds 10-14 days per Asia-Europe voyage and raises freight and insurance costs.
- **Safety at sea:** when most big ships are dark, collision avoidance, rescue and identification fail, and neutral ships risk being mistaken for targets.
- **Shadow fleets:** ships registered in flags often linked to sanctions evasion are about three times over-represented in war zones.

For India

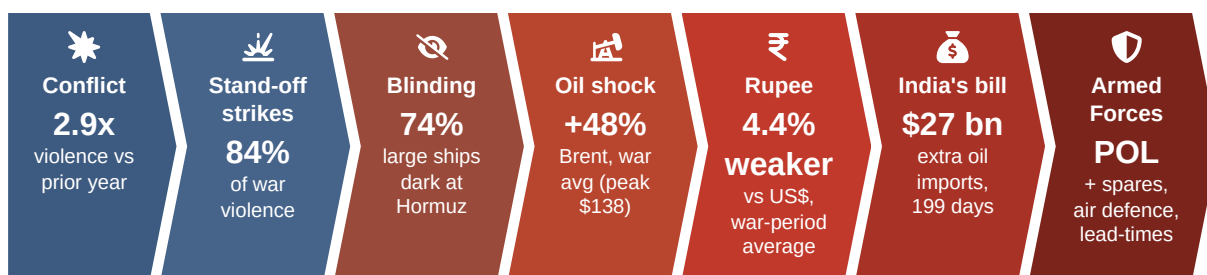


Figure 7.1 : How the shock travels from the strait to India's import bill and the Armed Forces

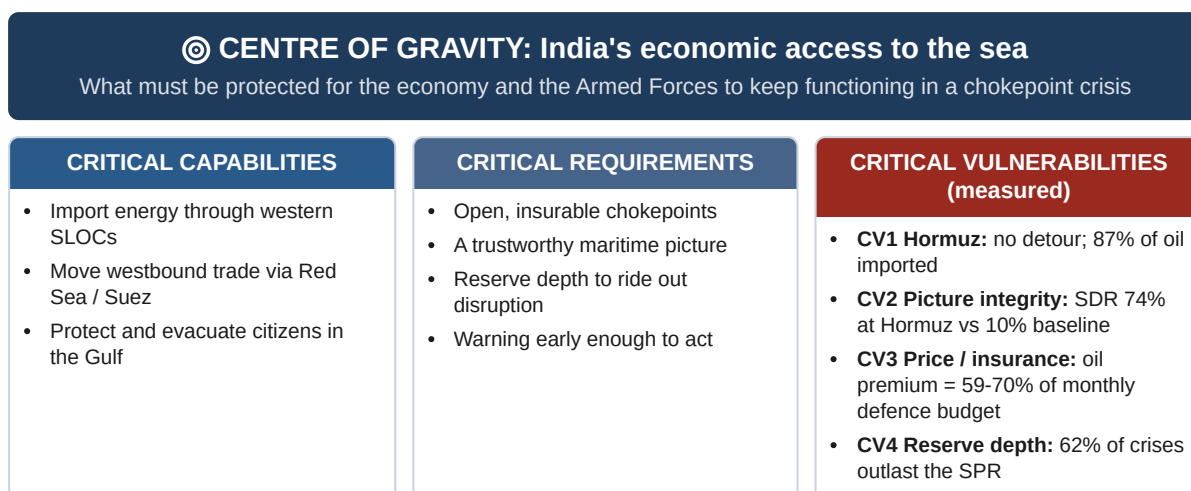


Figure 7.2 : India's centre of gravity at sea, with the vulnerabilities this study measured

- **Energy:** 87% of oil is imported, much of it through Hormuz, which has no detour. Only buffers and other suppliers help.
- **Trade:** the Red Sea diversion lengthens and raises the cost of India's westbound exports.
- **Connectivity:** the Chabahar corridor and IMEC run through the theatres studied.
- **People:** about nine million Indians live in the Gulf states, where violence rose 124-fold. At least 33 Indian-flag ships were inside the Gulf war zone.
- **Home waters:** India's own seas hold thousands of small boats without beacons, so coastal security cannot rely on AIS.

SO WHAT

For the world, chokepoint war is a recurring cost and a hazard at sea. For India the exposure is concentrated in oil, westbound trade, connectivity projects and nine million citizens in the Gulf.

Next: And for the Armed Forces?

8. WHAT IT MEANS FOR THE INDIAN ARMED FORCES

PART VII - WHAT IT MEANS FOR THE ARMED FORCES

THE STORY SO FAR

The same shocks that hit India's economy reach its Armed Forces through fuel, spares, threats to bases and the need to protect citizens abroad.

INDIAN NAVY

- Escort in a picture where 61% of large ships are dark
- Ship air defence and counter-drone magazines for long campaigns
- Coastal security against thousands of non-AIS small craft
- Shadow-fleet oil-spill response in the EEZ

INDIAN AIR FORCE

- Hardened, dispersed air bases and fuel farms vs stand-off strikes
- Air corridors to West Asia closed; longer airlift
- GNSS-resilient avionics and weapons (NavIC, anti-jam)
- Airlift for NEO of ~9 million citizens in GCC

INDIAN ARMY

- Layered AD / counter-UAS for depots, railheads, POL points
- GNSS-denied fallback for drones, loitering munitions, fires
- Western front: instability in Sistan-Baluchestan / Makran
- Spares lead-times for imported fleets lengthen

JOINT (HQ IDS): POL / aviation fuel / spares War Wastage Reserves sized on the survival curve (35-45 days) · ~US\$ 15.6 bn national cost per US\$10/bbl-year · tri-service Data Analytics Cell running the I&W matrix weekly · GCC NEO at execution readiness

Figure 8.1 : Impact on the Indian Navy, Air Force and Army

Prescriptive stock-cover curve from observed disruption durations

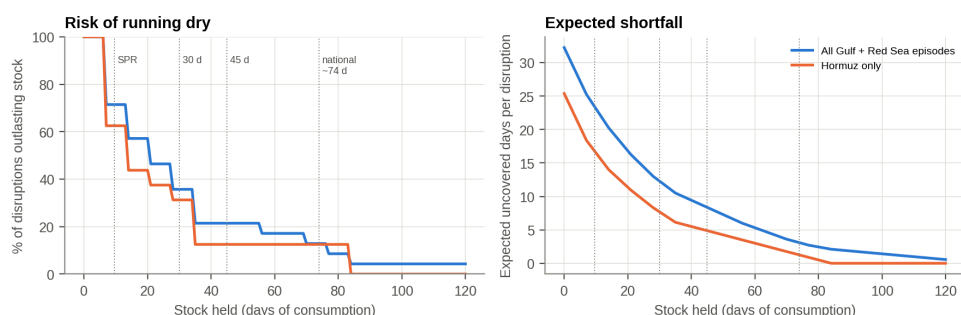


Figure 8.2 : How much stock is enough? Risk of running dry and average days short, by days of stock held

What this shows: Left: the chance that a crisis outlasts a given stock. Right: the average days short. Beyond about 35-45 days, extra stock buys little.

Table 8.1 : Planning scenarios for a Hormuz or Red Sea disruption

Scenario	Duration	P(disruption lasts at least this long)	Days beyond national cover (74 d)	Days beyond 30 d / 45 d Service holdings
S1 Short disruption	2 weeks	71%	0	0 / 0
S2 Sustained disruption	6 weeks	21%	0	12 / 0
S3 Prolonged disruption	12 weeks	9%	10	54 / 39
S4 Campaign-length disruption	26 weeks	< 5% (beyond the longest observed episode)	108	152 / 137

A six-week crisis is the planning case: roughly 1 in 5 crises last that long. It leaves a 30-day holding 12 days short, while 45 days covers it fully. Beyond about 35 days each extra day of stock buys little. Hence 35-45 days for fuel, aviation fuel and critical spares.

SO WHAT

Fuel and spares stocks sized on evidence, rear areas defended against drones and missiles, navigation that does not depend on GPS, and a joint watch on the chokepoints.

Next: What should we do, and in what order?

9. WAY FORWARD

PART VIII - WHAT WE SHOULD DO

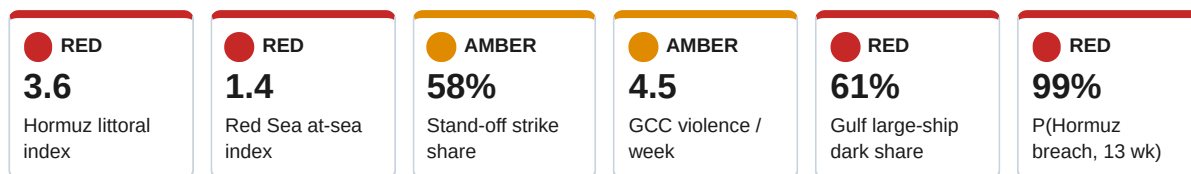
THE STORY SO FAR

The story has shown what happens, how long it lasts, what it costs and whom it hurts. Now: what to do, what to watch, and in what order.

Table 9.1 : Eleven staffed actions, each traced to the evidence

Action	Lead	Timeline	Measure of effectiveness
Maritime Picture Assurance as a named joint warning function; SDR and DARKWATCH as weekly products.	HQ IDS with IFC-IOR	4 months	A principal acts on a DARKWATCH warning
Standing rule: no AIS-derived traffic count for a contested corridor without a radar cross-check.	HQ IDS	Immediate	Rule issued; cited in assessments
Fuse national satellite radar (NISAR, EOS-04) with AIS to flag every dark big ship automatically.	Navy (IFC-IOR), DSA	6-18 months	Dark hulls cued within 24 h of a pass
Treat AIS identity as untrusted: automatic checks for cloned or fake identities.	Navy (IFC-IOR)	6 months	Checks shared with ports and insurers
Know the small-boat population in India's seas: transponders, coastal radar, satellite radar.	Navy, Coast Guard, MoFAH	18 months	Share of small craft identifiable
Size strategic oil and LPG reserves on crisis-duration evidence; diversify away from Hormuz.	MoPNG with MEA	12 months	Cover set on the survival curve
Re-set fuel, aviation-fuel and critical-spares War Wastage Reserves for all three Services at 35-45 days.	HQ IDS (Log)	6 months	Days of cover within the band
Air defence and counter-drone cover for depots, railheads, air bases and fuel points; GPS-denied fallback.	Army, IAF	3 years, rolling	Critical nodes covered; drills done
Keep the Gulf evacuation plan (Navy sealift, IAF airlift, Army reception) ready while signals are Amber or Red.	HQ IDS with MEA	3 months	Time to first lift from alert
Indigenise first the items that come through Hormuz or the Red Sea with long lead-times.	DDP, Services	Rolling	Exposure-weighted list adopted

Action	Lead	Timeline	Measure of effectiveness
Archive corridor satellite imagery and write up what shipping did after every chokepoint crisis.	HQ IDS, CDM	After each crisis	Report within 90 days



Out-of-sample test. On 26 Jun 2026 the market read calm: Brent at **\$70**, back to its pre-war level. The matrix read **4 Red**: hold buffers. Brent then rose **+86%** to **\$131** by 15 Sep 2026. The data called what the market missed.

Figure 9.1 : DARKWATCH: the six-signal watch-list at 27 June 2026, and what happened next

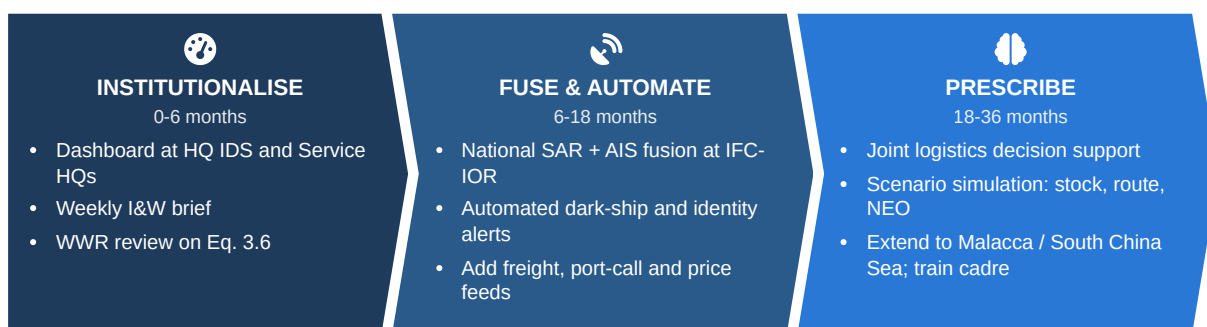


Figure 9.2 : Three-phase plan

SO WHAT

Eleven staffed actions, the DARKWATCH watch-list and a three-phase plan that starts at no new cost and stops if it does not help a decision.

10. CONCLUSION

EPILOGUE

THE STORY SO FAR

We end where we began, with the commander's questions, now answered with evidence.

- Visibility is the first casualty.** 74% of large ships dark at Hormuz; the effect is robust, graded and replicated.
- Detour decides the response.** No detour means blinding and a price shock; a detour means diversion and higher freight.
- Disruption is a duration problem.** 62% outlast the SPR; returns to stock flatten at ~35-45 days.
- Stand-off weapons erase the rear area.** 84% of war violence; energy and logistics nodes far from the front were hit.
- Data beats the market as a warning.** The June I&W call was Red at \$70 Brent; oil then rose 86%.

Figure 10.1 : Five takeaways

The first casualty of war at a chokepoint is visibility. Physical disruption follows, lasts longer than intuition suggests, and reaches India's energy, trade, citizens and military sustainment. The same data that measured it can warn of it. Turned into a standing weekly watch, the tools built here let India and its Armed Forces decide on evidence rather than precedent. That is the aim of this Datathon.

WHERE TO FIND THE DETAIL

The full technical report (method, all charts and tables, robustness tests, references) and the Power BI dashboard accompany this edition. Every number here is reproduced there, and can be regenerated from the raw data with one command.

ANNEX A - DATA ANALYTICS IN PLAIN WORDS

This report uses a number of data-analytics tools. None of them needs to be understood in detail to follow the story: every chapter opens with **The story so far**, closes with **So what**, and every chart carries a one-line **What this shows**. The terms below are explained in plain words for readers who want them.

Term	What it means, in plain words
AIS (Automatic Identification System)	A ship's identity beacon. It broadcasts name, position and course. It works like IFF for merchant ships, and a ship can switch it off.
AIS-dark ship	A ship that satellite radar can see but that is not transmitting its identity: a vehicle driving at night with its lights off.
SAR (satellite radar)	A radar satellite that sees ships day and night, through cloud, whether or not they transmit.
Chokepoint	A narrow sea passage that ships cannot easily avoid, such as Hormuz or Bab-el-Mandeb: a defile on the sea lanes.
Relative risk / 'x times more likely'	How much more often something happens in one group than another. A relative risk of 4.7 means 4.7 times as often.
p-value	The chance that a result is a fluke. Below 0.05 means less than 1 in 20. Most results here are far smaller, i.e. almost certainly real.
Confidence interval	The range in which the true value probably lies, like the beaten zone of a weapon rather than a single point.
Change-point detection	A method that finds the dates when the 'battle rhythm' changed, without being told what to look for.
Conflict intensity index (CCII)	A weekly thermometer of violence around each sea route, built from open conflict data.
Forecast (ARIMA)	Projects the thermometer forward a few months, with a band showing the likely range.
Survival analysis (Kaplan-Meier, Cox)	Measures how long crises last and what makes them last longer. It is the same tool used for time between equipment failures.
Hot-spot analysis (Getis-Ord Gi*)	Finds places where dark ships bunch together more than chance would allow, like an incident-density overlay.
Cluster finding (DBSCAN)	Automatically finds groups of ships close together, such as tanker queues at an anchorage.
Machine-learning model	A program that learns a rule from 62,000 examples. It was tested only on seas it had never seen, like testing a soldier on unfamiliar ground.
Model score (ROC-AUC)	How well a model separates dark ships from visible ones: 0.5 is a coin toss, 1.0 is perfect.
Bootstrap	Re-running the analysis 2,000 times on re-shuffled satellite pictures to check that the answer does not change.
Dose-response	The stronger the exposure, the stronger the effect. Here: the closer to the fighting, the more ships go dark.

Term	What it means, in plain words
Competing hypotheses	Standard intelligence tradecraft: list every possible explanation and keep the one the evidence does not contradict.
Event study	Tracking a price day by day after a shock, to see how big and how lasting the effect was.
Granger test	Checks whether A reliably happens before B, i.e. whether A could serve as a warning of B.
Expected shortfall	The average number of days a stock would run short in a crisis. It is used here to size fuel and spares reserves.
I&W (indicators and warnings)	A short watch-list of measurable signals with thresholds that trigger pre-agreed actions.
Strategic Dark Ratio (SDR)	Of the big ships (100 m and over) the radar sees in one satellite image, the share not transmitting their identity. Peaceful seas: about 10%.
Maritime Picture Assurance	Checking, as a standing duty, that the maritime picture used for decisions still represents the ships actually on the water.
DARKWATCH	The six-signal watch-list built in this study, with Amber and Red thresholds and a pre-agreed action for each.
Stasis index	The share of ships seen in the same 100 m square on two different days: a ship under way cannot do that, a ship at anchor can.
Difference-in-differences	Did the Gulf change more than everywhere else over the same fortnight? Compares the change, not just the level.
Estimative language	'Almost certain' (over 95%), 'highly likely' (80-95%), 'likely' (55-80%): the standard way intelligence expresses probability.

ANNEX B - TECHNICAL SUMMARY FOR ASSESSORS

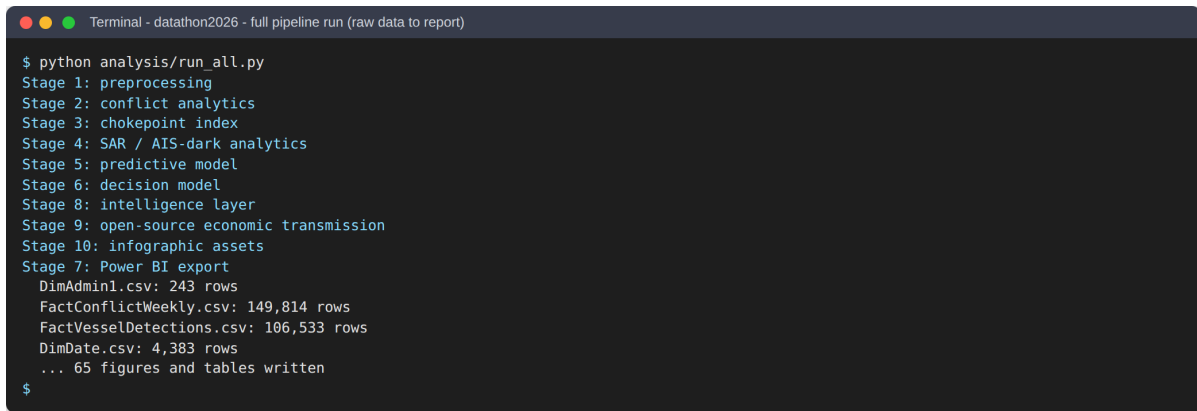
For assessors: every technique used, the plain question it answers, the key result, and where the full method, tests and tables appear in the accompanying technical report. All results are regenerated from raw data by one command, and the pipeline is deterministic (two consecutive runs give identical outputs).

Table B.1 : Techniques, questions and results

Technique	Commander's question	Key result	Full report
Validation, cleaning, feature engineering; BallTree haversine spatial join	Is the evidence sound?	149,814 conflict rows; 106,533 ships; 724 low-confidence detections dropped	Ch 4
Change-point detection (PELT, L2 cost); Mann-Whitney U	When did the battle rhythm change?	7 Oct 2023, 28 Feb 2026, 11 Apr 2026; war regime 2.9x ($p = 4e-05$)	5.1
Chokepoint Conflict Intensity Index; ARIMA with 12-week back-test and 4,000 simulations	Is the sea lane under stress, and will it stay so?	Hormuz littoral 188x baseline; 99% of simulated paths breach by Sep 2026	6.1-6.2
Survival analysis: Kaplan-Meier, Cox proportional hazards, log-rank	How long do crises last?	Median 2.5 wk; 62% exceed SPR; 11% exceed 74 days	6.3
Wilson intervals, chi-square, relative risk, odds ratio	Are war-zone ships really darker?	RR 4.7, OR 8.0, $p < 10^{-300}$	7.1
Getis-Ord Gi* local hot-spot statistic	Where do dark ships cluster?	43 significant cells, Qatar-Hormuz belt	7.2
DBSCAN density clustering (haversine)	Where do they wait?	36 clusters; Dubai 97% dark	7.3
Logistic regression; gradient-boosted trees with monotonic constraints; leave-one-region-out CV	Can we forecast darkness?	ROC-AUC 0.69 on unseen seas (random CV 0.74, reported as leakage)	Ch 8
Event study; Granger causality; pass-through regression; import-bill model	What did it cost, and do prices warn?	Brent +94% peak; no price lead ($p > 0.4$); -US\$27 bn to India	9.2-9.7
Scene-cluster bootstrap (8 specifications); dose-response logit; Analysis of Competing Hypotheses	How sure are we?	RR 3.9-6.4; 76% to 43% with distance; switch-off 0 inconsistencies	10.4-10.6
Expected shortfall from the survival curve; scenario matrix	How much stock is enough?	Returns flatten at ~35 days; 45 days covers the 6-week case	13.5-13.6
Presence vs identity on a common footprint; stasis index against a crowding null	Did ships leave or go silent?	AIS -83% vs radar -33%; dark hulls held in place ($z = 5.3$)	7.8
Scene-level difference-in-differences with 5,000 permutations; all-dark and day/night checks	Did the Gulf change more than elsewhere?	+5.7 pts/day vs controls ($p = 0.008$); RR 7.5 without all-dark passes	10.4
Flag-retention shift (chi-square); rule-based signature assignment	Who kept transmitting; what kind of event is it?	Gulf flags 11% to 20% of lit fleet; four signatures	7.9-7.10
Indicators & warnings matrix; out-of-sample test	Would we have been warned?	4 Red / 2 Amber at \$70 Brent; oil then +86%	9.8, 14.2

ANNEX C - SOFTWARE USED AND SCREENSHOTS

Stack: Python 3.11 (pandas, NumPy, SciPy, statsmodels, scikit-learn, ruptures, Matplotlib/Basemap) for analysis; headless Chromium for the report; Microsoft Power BI Desktop for the dashboard (.pbix) built on the exported star schema (FactConflictWeekly, FactVesselDetections, DimAdmin1, DimDate and the analysis tables).

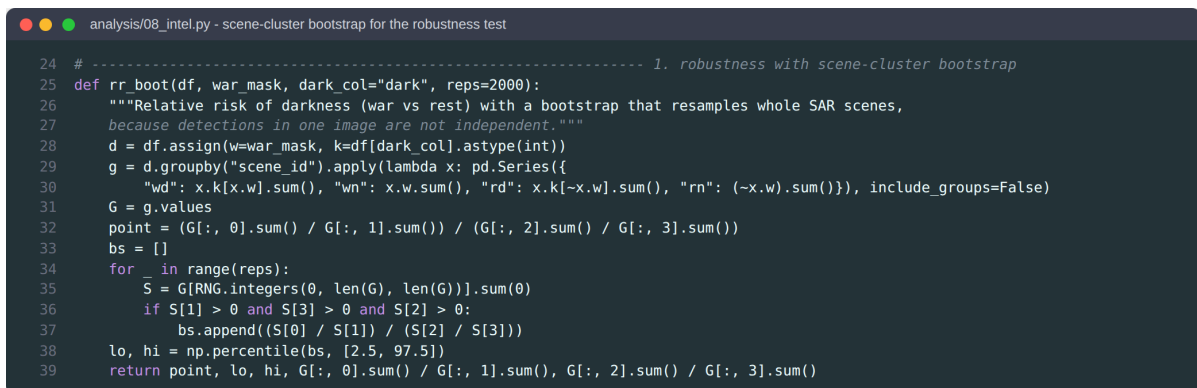


```

$ python analysis/run_all.py
Stage 1: preprocessing
Stage 2: conflict analytics
Stage 3: chokepoint index
Stage 4: SAR / AIS-dark analytics
Stage 5: predictive model
Stage 6: decision model
Stage 8: intelligence layer
Stage 9: open-source economic transmission
Stage 10: infographic assets
Stage 7: Power BI export
DimAdmin1.csv: 243 rows
FactConflictWeekly.csv: 149,814 rows
FactVesselDetections.csv: 106,533 rows
DimDate.csv: 4,383 rows
... 65 figures and tables written
$

```

Figure C.1 : Screenshot: the complete pipeline run, raw data to every chart, table and dashboard table



```

24 # ----- 1. robustness with scene-cluster bootstrap
25 def rr_boot(df, war_mask, dark_col="dark", reps=2000):
26     """Relative risk of darkness (war vs rest) with a bootstrap that resamples whole SAR scenes,
27     because detections in one image are not independent."""
28     d = df.assign(w=war_mask, k=df[dark_col].astype(int))
29     g = d.groupby("scene_id").apply(lambda x: pd.Series({
30         "wd": x.k[x.w].sum(), "wn": x.w.sum(), "rd": x.k[~x.w].sum(), "rn": (~x.w).sum()}), include_groups=False)
31     G = g.values
32     point = (G[:, 0].sum() / G[:, 1].sum()) / (G[:, 2].sum() / G[:, 3].sum())
33     bs = []
34     for _ in range(reps):
35         S = G[RNG.integers(0, len(G), len(G))].sum(0)
36         if S[1] > 0 and S[3] > 0 and S[2] > 0:
37             bs.append((S[0] / S[1]) / (S[2] / S[3]))
38     lo, hi = np.percentile(bs, [2.5, 97.5])
39     return point, lo, hi, G[:, 0].sum() / G[:, 1].sum(), G[:, 2].sum() / G[:, 3].sum()

```

Figure C.2 : Screenshot: code excerpt, scene-cluster bootstrap behind the robustness test

[Insert screenshot: Power BI page 1 - Conflict Pulse / page 2 - Chokepoint Watch]

Figure C.3 : Screenshot: Power BI dashboard, conflict and chokepoint pages

[Insert screenshot: Power BI page 3 - Dark Ships / page 4 - Decision]

Figure C.4 : Screenshot: Power BI dashboard, dark-ship and decision pages

ANNEX D - KEY REFERENCES AND DATA SOURCES

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8. Roberts, D.R. et al. (2017). Cross-validation strategies for structured data. *Ecography* 40, 913-929.
9. Open data: EIA Brent/WTI spot prices (github.com/datasets/oil-prices); Federal Reserve H.10 INR/US\$ (github.com/datasets/exchange-rates); Energy Institute Statistical Review via Our World in Data (github.com/owid/energy-data); Natural Earth ports. Icons: Font Awesome Free (CC BY 4.0).
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